

*"I think,
therefore I am"*

The French mathematician René Descartes (1596–1650) used logic to prove that there was only one thing he could be completely sure of – his own existence. He summed it up in Latin: *Cogito, ergo sum* ("I think, therefore I am"). Descartes' main claim to fame was the invention of Cartesian coordinates. He's less well known for doing all his greatest work in bed.



THREE DOORS

You're on a TV game show. The compere shows you three closed doors and tells you there's a shiny black sports car behind one of them. If you choose the right door, you win it. You pick a door at random. The compere, who knows where the car is, then opens another door and shows you an empty room. He asks if you want to change your mind. Should you?

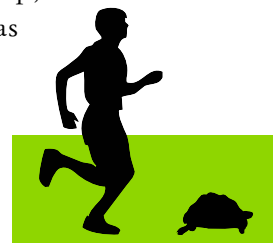
THREE HATS

Three sisters, A, B, and C, are wearing hats, which they know are either black or white but not all are white. A can see the hats of B and C; B can see the hats of A and C; C is blindfolded. Each is asked in turn if they know the colour of their own hat. The answers are: A: "No." B: "No." C: "Yes." What colour is C's hat and how does she know?

ZENO'S PARADOX

A Greek philosopher called Zeno thought up a paradox involving the idea of infinity. A man called Achilles challenges a tortoise to a race. Suppose Achilles can run ten times faster than the tortoise, but he gives the tortoise a 10 metre head-start. When Achilles has run 10 metres, the tortoise has run 1 metre and is still in the lead. When Achilles has covered that 1 metre, the tortoise has moved another tenth of a metre forward. Each time Achilles tries to catch up, the tortoise has gone a bit further still. This can continue forever, the tortoise moving forward by ever smaller increments. So it seems logical that Achilles can never overtake the tortoise – yet common sense tells us that anyone can overtake a tortoise in a race!

The puzzle baffled the Greeks because they didn't understand the idea of infinity. They thought an infinite number of values, however tiny, must add up to an infinite amount. The problem wasn't fully solved until the 1600s, when a Scottish mathematician, James Gregory, showed that an infinite number of ever-decreasing values can add up to a *finite* amount.



The sentence on the left is false.